

E075

Special Topics in Software Engineering and AI

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A developer asks an AI coding assistant, trained on a very large amount of public code, how to parse dates. It suggests calling a library function that the developer has never seen. The developer's team ships every Tuesday. What should the developer do?

- A. Never use AI coding tools.
- B. Use it, because a model trained on so much code is reliable.
- C. Check the library's official docs and cover the code with tests before using it.
- D. Raise the model's temperature and ask again.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A company of 3,000 employees wants an internal chatbot that answers questions from HR policies, which change every month. The chatbot's logo was designed by an intern. Which approach fits best?

- A. Fine-tune the model on this month's policies.
- B. Have a professional redesign the logo so employees trust the answers.
- C. Retrieval-augmented generation over the current policy documents.
- D. Rely on the base model's general knowledge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A team needs a model to always reply in a strict internal JSON format and a consistent tone, and prompting alone has been unreliable. The underlying facts do not change. Which approach fits best?

- A. A vector database alone.
- B. Retrieval-augmented generation.
- C. Fine-tuning on examples of the desired output.
- D. A larger context window.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team built, in six weeks, an AI assistant that reads incoming customer emails and can issue refunds. Leaders worry that text inside an email could steer the assistant into unwanted actions. Which design choice best reduces that risk?

- A. Encrypt the emails at rest.
- B. Rebuild the assistant more slowly.
- C. Add a line to the system prompt telling the model to ignore bad instructions.
- D. Treat email content as data only, limit the refund tool's permissions, and require human approval above a set amount.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An AI bug detector, running on a team's 400,000-line code base, flags 50 functions as buggy. Review finds 40 of them really are buggy. The code base has 80 buggy functions in total. What is the detector's precision?

- A. 20%
- B. 80%
- C. 62.5%
- D. 50%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Using the same numbers (50 flagged, 40 correct, 80 real bugs), what is the detector's recall?

- A. 40%
- B. 50%
- C. 100%, since every flag was reviewed
- D. 80%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A résumé-screening model is trained on ten years of a firm's past hiring decisions, which strongly favored one group. The firm has offices in four states. The team removes the gender column and plans to launch. What is the main concern?

- A. The model can still learn the old bias through proxy features; audit outcomes across groups before launch.
- B. AI models are objective, so bias is not possible.
- C. The model should be trained separately for each state.
- D. None; removing the gender column removes bias.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A four-person startup, funded by a regional investor, is building its first product and expects requirements to change weekly. An engineer proposes splitting it into 15 microservices from day one. What is the better starting point?

- A. 15 microservices, because they are always more scalable.
- B. A well-structured modular monolith, splitting services later if needed.
- C. Ask the investor to choose the architecture.
- D. One microservice per developer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A large retailer has 12 product teams. One checkout component needs to scale separately during holiday peaks, and teams keep blocking each other's releases. Which architecture fits best?

- A. A serverless function for every line of code.
- B. A single monolith with one release train.
- C. Buy larger servers for the whole app.
- D. Microservices with clear service ownership.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A fraud model's inputs look the same as last year, but new regulations changed which transactions count as fraud, and accuracy has dropped. The model runs on a cluster in two regions. What is happening?

- A. Concept drift; retrain with newly labeled data.
- B. A cluster region failure.
- C. Overfitting during training.
- D. Data drift.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A recommendation model was trained mostly on users aged 30–50. After a campaign, most new users are 18–24, and results get worse. What kind of drift is this?

- A. Label leakage
- B. Concept drift
- C. No drift; models stay accurate once deployed.
- D. Data drift

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A lender plans to let an AI model approve or deny loans with no human review. The model scored well in testing and runs on the newest GPUs. What is the most responsible design?

- A. Keep human review for denials and borderline cases, explain outcomes, and offer appeals.
- B. Upgrade the GPUs again.
- C. Publish the model's source code and skip review.
- D. Automate fully, since the model tested well.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A developer asks an AI coding assistant, ~~trained on a very large amount of public code~~, how to parse dates. It suggests calling a library function that the developer has never seen. ~~The developer's team ships every Tuesday.~~ What should the developer do?

Target word: handling unverified AI output **Decides:** The function may not exist; output must be checked.

A	Absolute language	They are useful when output is verified.
B	Camouflage	Training size does not prevent hallucination.
C	Correct	Verifies against a trusted source.
D	Inverted operation	More randomness makes output less reliable.

2. Answer C Concept 2

A company ~~of 3,000 employees~~ wants an internal chatbot that answers questions from HR policies, which change every month. ~~The chatbot's logo was designed by an intern.~~ Which approach fits best?

Target word: keep answers current **Decides:** The facts change monthly.

A	Confusion set	Must retrain every month and still may misremember facts.
B	Camouflage	Branding does not supply knowledge.
C	Correct	Fetches the latest text at query time.
D	Dropped element	It has never seen the company's policies.

3. Answer C Concept 2 · transfer

A team needs a model to always reply in a strict internal JSON format and a consistent tone, and prompting alone has been unreliable. The underlying facts do not change. Which approach fits best?

Target word: change behavior, not knowledge **Decides:** The need is consistent format and style.

A	Confusion set	Stores embeddings; does not change output format.
B	Inverted operation	Adds facts, not a consistent output style.
C	Correct	Shapes the model's behavior.
D	Adjacent category	Fits more input, but does not train the format.

4. Answer D Concept 3

A team built, ~~in six weeks~~, an AI assistant that reads incoming customer emails and can issue refunds. Leaders worry that text inside an email could steer the assistant into unwanted actions. Which design choice best reduces that risk?

Target word: defensive control for injected instructions **Decides:** Outside content is untrusted and the tool is high impact.

A	Adjacent category	Protects storage, not what the model does with content.
B	Camouflage	Build time is not the control.
C	Dropped element	Helpful but not sufficient alone.
D	Correct	Layered controls: least privilege plus human review.

5. Answer B Concept 4

An AI bug detector, running on a team's 400,000 line code base, flags 50 functions as buggy. Review finds 40 of them really are buggy. The code base has 80 buggy functions in total. What is the detector's precision?

Target word: precision **Decides:** Correct flags ÷ all flags.

A	Inverted operation	10 / 50 is the false-flag share.
B	Correct	40 / 50.
C	Adjacent category	50 / 80 compares flags to total bugs.
D	Confusion set	40 / 80 is recall.

6. Answer B Concept 4 · transfer

Using the same numbers (50 flagged, 40 correct, 80 real bugs), what is the detector's recall?

Target word: recall **Decides:** Correct flags ÷ all real bugs.

A	Wrong stem number	The count of correct flags, read as a percent.
B	Correct	40 / 80.
C	Absolute language	Reviewing flags does not find missed bugs.
D	Inverted operation	That is precision.

7. Answer A Concept 5

A résumé-screening model is trained on ten years of a firm's past hiring decisions, which strongly favored one group. The firm has offices in four states. The team removes the gender column and plans to launch. What is the main concern?

Target word: bias persists through proxies **Decides:** Historical decisions carried bias; other features can stand in for the removed one.

A	Correct	Removing one column does not remove learned patterns.
B	Absolute language	Models reflect their training data.
C	Camouflage	Office locations do not address the bias.
D	Dropped element	Ignores proxies like school or activities.

8. Answer B Concept 6

A four-person startup, funded by a regional investor, is building its first product and expects requirements to change weekly. An engineer proposes splitting it into 15 microservices from day one. What is the better starting point?

Target word: architecture fit **Decides:** Small team, uncertain domain.

A	Absolute language	Scaling benefits come with heavy costs a small team cannot absorb.
B	Correct	Less operational overhead while the domain settles.
C	Camouflage	Funding source does not decide the design.
D	Adjacent category	Service boundaries should follow the domain, not headcount.

9. Answer D Concept 6 · transfer

A large retailer has 12 product teams. One checkout component needs to scale separately during holiday peaks, and teams keep blocking each other's releases. Which architecture fits best?

Target word: when microservices pay off **Decides:** Many teams and independent scaling needs.

A	Confusion set	Extreme fragmentation, not a service design.
B	Inverted operation	Keeps the release blocking.
C	Dropped element	Does not fix team coupling.
D	Correct	Independent deployment and scaling.

10. Answer A Concept 7

A fraud model's inputs look the same as last year, but new regulations changed which transactions count as fraud, and accuracy has dropped. ~~The model runs on a cluster in two regions.~~ What is happening?

Target word: name the drift **Decides:** The input-to-label relationship changed.

A	Correct	Same inputs, new meaning of the label.
B	Camouflage	Infrastructure does not change label meaning.
C	Adjacent category	A training-time issue, not a change over time.
D	Confusion set	The inputs did not change.

11. Answer D Concept 7 · transfer

A recommendation model was trained mostly on users aged 30–50. After a campaign, most new users are 18–24, and results get worse. What kind of drift is this?

Target word: name the drift **Decides:** The input distribution shifted.

A	Confusion set	A training-data flaw, not a population change.
B	Inverted operation	Would require the input-to-label relationship to change.
C	Absolute language	Production data changes over time.
D	Correct	Input populations changed.

12. Answer A Concept 8

A lender plans to let an AI model approve or deny loans with no human review. The model scored well in testing ~~and runs on the newest GPUs.~~ What is the most responsible design?

Target word: oversight for high-stakes decisions **Decides:** Loan decisions significantly affect people.

A	Correct	Human oversight and recourse for high-stakes decisions.
B	Camouflage	Hardware does not create accountability.
C	Adjacent category	Transparency alone is not oversight or appeal.
D	Inverted operation	Good test scores do not remove the need for oversight.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Verify generative AI output	Large language models can produce fluent but false output (hallucinations), so treat generated code and facts as drafts to check against sources and tests.	Trusting output because it sounds confident or the model is large.	An AI suggestion that is used without checking.
2	RAG vs. fine-tuning	Retrieval-augmented generation supplies current or private facts at query time; fine-tuning changes a model's behavior, style, or format.	Fine-tuning to teach facts that change often.	Changing knowledge vs. a consistent behavior.
3	Prompt injection as a design risk	Treat any content an AI system reads as untrusted data, give its tools least privilege, and require human approval for high-impact actions.	Relying on a system prompt that says "ignore bad instructions."	An AI agent that reads outside content and can take actions.
4	Precision and recall	Precision = true positives ÷ everything flagged; recall = true positives ÷ everything that should have been flagged.	Swapping the two denominators.	Counts of flagged items, correct flags, and total real cases.
5	Bias from training data	Models learn patterns in historical data, including unfair ones; mitigate by auditing outcomes across groups, fixing data, and keeping human review.	Deleting a protected attribute and assuming bias is gone.	A model trained on past human decisions.
6	Monolith vs. microservices	A modular monolith suits small teams and new, uncertain domains; microservices pay off when many teams need independent deployment and scaling.	Choosing microservices by default because they are modern.	Team size, number of teams, and scaling needs.
7	Data drift vs. concept drift	Data drift is a change in input distributions; concept drift is a change in the relationship between inputs and the correct output. Both call for monitoring and retraining.	Assuming a model stays accurate once deployed.	Production performance declining over time.
8	Human oversight for high-stakes AI	Where decisions significantly affect people, keep a human in the loop, explain outcomes, and provide a way to appeal.	Full automation because the model scored well in testing.	Loans, hiring, health, or legal decisions made by AI.

Course scope (public description): This course explores modern topics, challenges, and emerging solutions in software engineering and artificial intelligence. Students analyze contemporary frameworks, methodologies, and technologies, and apply them to solve real- world problems. Through applied projects, learners design and develop software solutions, evaluate the use of emerging technologies, and assess technical, ethical, and professional implications. The course culminates in a completed application that demonstrates the ability to analyze, design, and implement a modern software solution.